

NATIONAL QUANTUM MISSION AND QUANTUM TECHNOLOGY

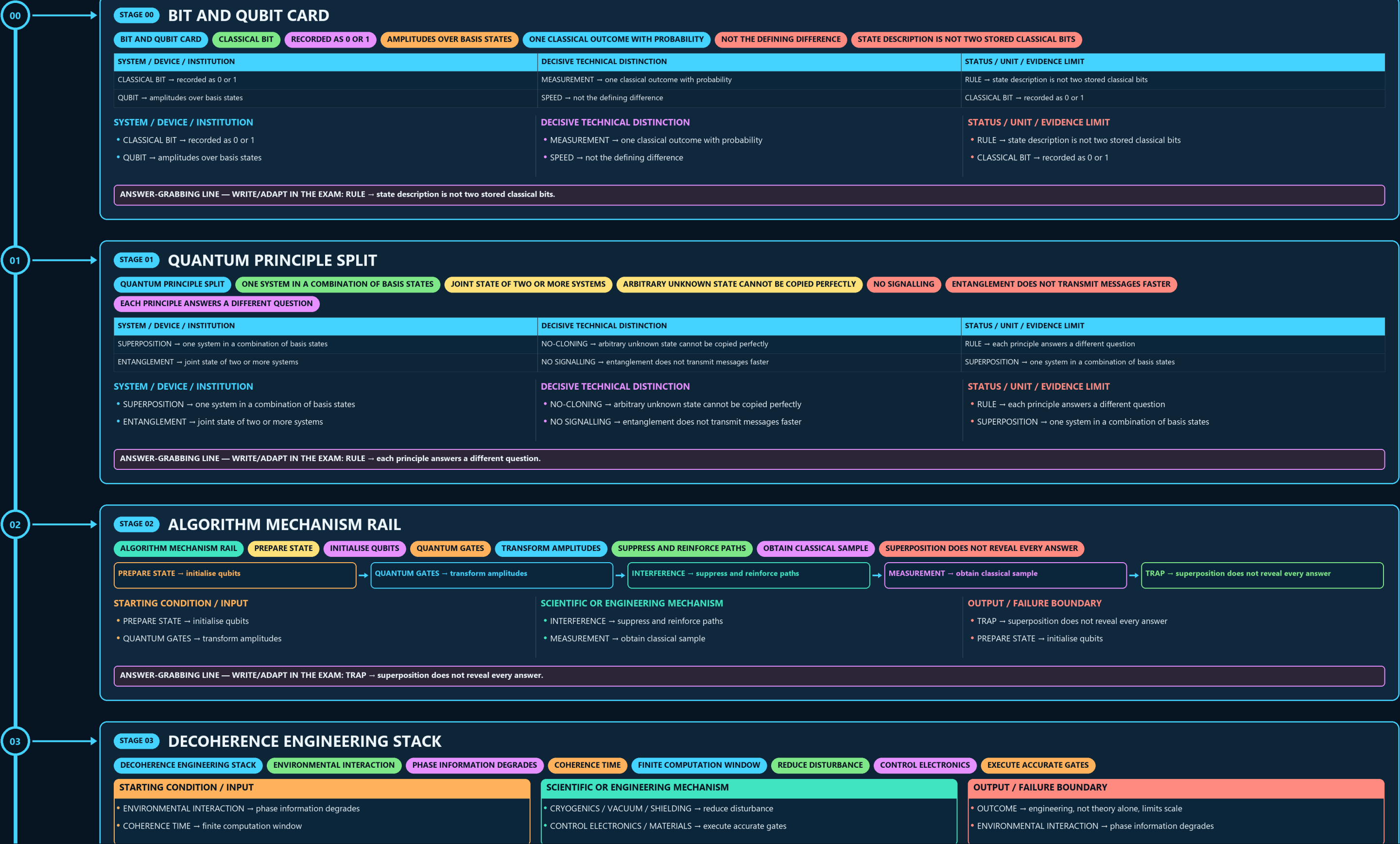
BIT AND QUBIT CARD → QUANTUM PRINCIPLE SPLIT → ALGORITHM MECHANISM RAIL → DECOHERENCE ENGINEERING STACK → PHYSICAL-TO-LOGICAL LADDER → FOUR TECHNOLOGY FAMILIES

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- PREPARE STATE → initialise qubits
- QUANTUM GATES → transform amplitudes

- INTERFERENCE → suppress and reinforce paths
- MEASUREMENT → obtain classical sample

- TRAP → superposition does not reveal every answer
- PREPARE STATE → initialise qubits

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → superposition does not reveal every answer.

03

STAGE 03

DECOHERENCE ENGINEERING STACK

DECOHERENCE ENGINEERING STACKENVIRONMENTAL INTERACTIONPHASE INFORMATION DEGRADES

COHERENCE TIMEFINITE COMPUTATION WINDOWREDUCE DISTURBANCECONTROL ELECTRONICSEXECUTE ACCURATE GATES

STARTING CONDITION / INPUT

- ENVIRONMENTAL INTERACTION → phase information degrades
- COHERENCE TIME → finite computation window

SCIENTIFIC OR ENGINEERING MECHANISM

- CRYOGENICS / VACUUM / SHIELDING → reduce disturbance
- CONTROL ELECTRONICS / MATERIALS → execute accurate gates

OUTPUT / FAILURE BOUNDARY

- OUTCOME → engineering, not theory alone, limits scale
- ENVIRONMENTAL INTERACTION → phase information degrades

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → engineering, not theory alone, limits scale.

04

STAGE 04

PHYSICAL-TO-LOGICAL LADDER

PHYSICAL-TO-LOGICAL LADDERPHYSICAL QUBITNOISY HARDWARE ELEMENT

ERROR CHARACTERISATIONMEASURE FAILURE CHANNELSERROR-CORRECTING CODEREDUNDANT ENCODINGLOGICAL QUBIT

PHYSICAL QUBIT → noisy hardware element

ERROR CHARACTERISATION → measure failure channels

ERROR-CORRECTING CODE → redundant encoding

LOGICAL QUBIT → protected information unit

STARTING CONDITION / INPUT

- PHYSICAL QUBIT → noisy hardware element
- ERROR CHARACTERISATION → measure failure channels

SCIENTIFIC OR ENGINEERING MECHANISM

- ERROR-CORRECTING CODE → redundant encoding
- LOGICAL QUBIT → protected information unit

OUTPUT / FAILURE BOUNDARY

- RULE → raw qubit count is not useful capability
- PHYSICAL QUBIT → noisy hardware element

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → raw qubit count is not useful capability.

05

STAGE 05

FOUR TECHNOLOGY FAMILIES

FOUR TECHNOLOGY FAMILIESSSELECTED INFORMATION-PROCESSING ALGORITHMSQUANTUM-STATE LINKS AND KEY DISTRIBUTIONPRECISION MEASUREMENTENABLING HARDWARE SUBSTRATE

QUANTUM TECHNOLOGY IS NOT COMPUTING ALONE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
COMPUTING → selected information-processing algorithms	SENSING / METROLOGY → precision measurement	TRAP → quantum technology is not computing alone
COMMUNICATION → quantum-state links and key distribution	MATERIALS / DEVICES → enabling hardware substrate	COMPUTING → selected information-processing algorithms

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
• COMPUTING → selected information-processing algorithms • COMMUNICATION → quantum-state links and key distribution	• SENSING / METROLOGY → precision measurement • MATERIALS / DEVICES → enabling hardware substrate	• TRAP → quantum technology is not computing alone • COMPUTING → selected information-processing algorithms

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → quantum technology is not computing alone.

06

STAGE 06

QKD SECURITY CHAIN

QKD SECURITY CHAINQUANTUM STATESDISTRIBUTE KEY MATERIALDISTURBANCE TESTREVEAL POSSIBLE INTERCEPTIONAUTHENTICATED ENDPOINTS

PREVENT IMPERSONATIONCLASSICAL ENCRYPTION

QUANTUM STATES → distribute key material

DISTURBANCE TEST → reveal possible interception

AUTHENTICATED ENDPOINTS → prevent impersonation

CLASSICAL ENCRYPTION → protects message content

RULE → QKD alone is not a complete secure network

STARTING CONDITION / INPUT

- QUANTUM STATES → distribute key material
- DISTURBANCE TEST → reveal possible interception

SCIENTIFIC OR ENGINEERING MECHANISM

- AUTHENTICATED ENDPOINTS → prevent impersonation
- CLASSICAL ENCRYPTION → protects message content

OUTPUT / FAILURE BOUNDARY

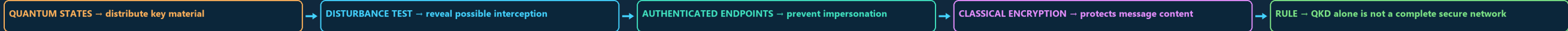
- RULE → QKD alone is not a complete secure network
- QUANTUM STATES → distribute key material

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → QKD alone is not a complete secure network.

06

STAGE 06 QKD SECURITY CHAIN

QKD SECURITY CHAIN QUANTUM STATES DISTRIBUTE KEY MATERIAL DISTURBANCE TEST REVEAL POSSIBLE INTERCEPTION AUTHENTICATED ENDPOINTS PREVENT IMPERSONATION CLASSICAL ENCRYPTION



STARTING CONDITION / INPUT

- QUANTUM STATES → distribute key material
- DISTURBANCE TEST → reveal possible interception

SCIENTIFIC OR ENGINEERING MECHANISM

- AUTHENTICATED ENDPOINTS → prevent impersonation
- CLASSICAL ENCRYPTION → protects message content

OUTPUT / FAILURE BOUNDARY

- RULE → QKD alone is not a complete secure network
- QUANTUM STATES → distribute key material

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → QKD alone is not a complete secure network.

07

STAGE 07 QKD VERSUS PQC

QKD VERSUS PQC QUANTUM HARDWARE AND PHYSICAL LINK CLASSICAL QUANTUM-RESISTANT ALGORITHMS QKD PURPOSE SPECIALISED KEY DISTRIBUTION PQC PURPOSE BROAD STANDARDS AND SOFTWARE MIGRATION COMPLEMENTARY TOOLS WITH DIFFERENT TIMELINES

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
QKD → quantum hardware and physical link	QKD PURPOSE → specialised key distribution	POLICY → complementary tools with different timelines
PQC → classical quantum-resistant algorithms	PQC PURPOSE → broad standards and software migration	QKD → quantum hardware and physical link

SYSTEM / DEVICE / INSTITUTION

- QKD → quantum hardware and physical link
- PQC → classical quantum-resistant algorithms

DECISIVE TECHNICAL DISTINCTION

- QKD PURPOSE → specialised key distribution
- PQC PURPOSE → broad standards and software migration

STATUS / UNIT / EVIDENCE LIMIT

- POLICY → complementary tools with different timelines
- QKD → quantum hardware and physical link

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PQC PURPOSE → broad standards and software migration.

08

STAGE 08 NQM INSTITUTION MAP

NQM INSTITUTION MAP MINISTRY OF SCIENCE AND TECHNOLOGY MINISTERIAL UMBRELLA IMPLEMENTING DEPARTMENT NATIONAL CAPABILITY MISSION FOUR VERTICAL ECOSYSTEM ANCHORS DEPARTMENT, MISSION AND HUB ARE DISTINCT



CONCEPT / ARCHITECTURE

- MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella
- DST → implementing department

MECHANISM / APPLICATION

- NQM → national capability mission
- T-HUBS → four vertical ecosystem anchors

READINESS / STATUS LIMIT

- RULE → department, mission and hub are distinct
- MINISTRY OF SCIENCE AND TECHNOLOGY → ministerial umbrella

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → department, mission and hub are distinct.

09

STAGE 09 FOUR T-HUB MAP

FOUR T-HUB MAP IISC BENGALURU QUANTUM COMPUTING IIT MADRAS WITH C-DOT QUANTUM COMMUNICATION IIT BOMBAY QUANTUM SENSING AND METROLOGY IIT DELHI

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
IISc BENGALURU → Quantum Computing	IIT BOMBAY → Quantum Sensing and Metrology	TRAP → hub creation is not operational capability
IIT MADRAS WITH C-DOT → Quantum Communication	IIT DELHI → Quantum Materials and Devices	IISc BENGALURU → Quantum Computing

SYSTEM / DEVICE / INSTITUTION

- IISc BENGALURU → Quantum Computing
- IIT MADRAS WITH C-DOT → Quantum Communication

DECISIVE TECHNICAL DISTINCTION

- IIT BOMBAY → Quantum Sensing and Metrology
- IIT DELHI → Quantum Materials and Devices

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → hub creation is not operational capability
- IISc BENGALURU → Quantum Computing

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → hub creation is not operational capability.

FOUR T-HUB MAP

FOUR T-HUB MAP

IISC BENGALURU

QUANTUM COMPUTING

IIT MADRAS WITH C-DOT

QUANTUM COMMUNICATION

IIT BOMBAY

QUANTUM SENSING AND METROLOGY

IIT DELHI

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
IISc BENGALURU → Quantum Computing	IIT BOMBAY → Quantum Sensing and Metrology	TRAP → hub creation is not operational capability
IIT MADRAS WITH C-DOT → Quantum Communication	IIT DELHI → Quantum Materials and Devices	IISc BENGALURU → Quantum Computing

SYSTEM / DEVICE / INSTITUTION

- IISc BENGALURU → Quantum Computing
- IIT MADRAS WITH C-DOT → Quantum Communication

DECISIVE TECHNICAL DISTINCTION

- IIT BOMBAY → Quantum Sensing and Metrology
- IIT DELHI → Quantum Materials and Devices

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → hub creation is not operational capability
- IISc BENGALURU → Quantum Computing

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → hub creation is not operational capability.

STAGE 10

READINESS LADDER

READINESS LADDER

CABINET APPROVAL

MISSION AUTHORITY AND OUTLAY

INTENDED FUTURE CAPABILITY

RESEARCH PROTOTYPE

BOUNDED DEMONSTRATION

VALIDATED SYSTEM

REPEATABLE PERFORMANCE EVIDENCE

CABINET APPROVAL → mission authority and outlay

TARGET → intended future capability

→

RESEARCH PROTOTYPE → bounded demonstration

→

VALIDATED SYSTEM → repeatable performance evidence

STARTING CONDITION / INPUT

- CABINET APPROVAL → mission authority and outlay
- TARGET → intended future capability

SCIENTIFIC OR ENGINEERING MECHANISM

- RESEARCH PROTOTYPE → bounded demonstration
- VALIDATED SYSTEM → repeatable performance evidence

OUTPUT / FAILURE BOUNDARY

- OPERATIONAL DEPLOYMENT → field use with sustainment
- CABINET APPROVAL → mission authority and outlay

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VALIDATED SYSTEM → repeatable performance evidence; OPERATIONAL DEPLOYMENT → field use with sustainment.

STAGE 11

QUANTUM CLAIM FIREWALL

QUANTUM CLAIM FIREWALL

QUBIT, SUPERPOSITION, ENTANGLEMENT AND DECOHERENCE

COMPUTING, COMMUNICATION, SENSING OR MATERIALS

PHYSICAL AND LOGICAL QUBITS; QKD AND PQC

APPROVAL, TARGET, PROTOTYPE, VALIDATION AND OPERATION

OUTLAY, QUBITS, DISTANCE AND STARTUP CLAIMS OFFICIALLY

DEFINE → qubit, superposition, entanglement and decoherence

→

CLASSIFY → computing, communication, sensing or materials

→

SEPARATE → physical and logical qubits; QKD and PQC

→

LABEL → approval, target, prototype, validation and operation

→

VERIFY → outlay, qubits, distance and startup claims officially

SYSTEM / DEVICE / INSTITUTION

- DEFINE → qubit, superposition, entanglement and decoherence
- CLASSIFY → computing, communication, sensing or materials

DECISIVE TECHNICAL DISTINCTION

- SEPARATE → physical and logical qubits; QKD and PQC
- LABEL → approval, target, prototype, validation and operation

STATUS / UNIT / EVIDENCE LIMIT

- VERIFY → outlay, qubits, distance and startup claims officially
- DEFINE → qubit, superposition, entanglement and decoherence

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → outlay, qubits, distance and startup claims officially.

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

QUANTUM CAPABILITY IS PATH-DEPENDENT

NATIONS THAT BUILD TALENT, FABRICATION KNOW-HOW AND STANDARDS EARLY

THE MOST BINDING CONSTRAINTS ARE OFTEN ENGINEERING AND ECOSYSTEM

A MISSION APPROACH IS JUSTIFIED BECAUSE FRAGMENTED PROJECTS ALONE

NATIONAL MISSION FRAMEWORK.

T-HUB AT IISC BENGALURU

QUANTUM COMPUTING.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: Quantum capability is path-dependent: nations that build talent, fabrication know-how and standards early can gain long-term advantages.
- CAUTION: The most binding constraints are often engineering and ecosystem constraints rather than pure theory.
- CAUTION: A mission approach is justified because fragmented projects alone may not create platform-level capability.
- NQM: national mission framework.

READINESS, UNIT OR STATUS LIMIT

- T-Hub at IISc Bengaluru: Quantum Computing.
- T-Hub at IIT Madras with C-DOT: Quantum Communication.
- T-Hub at IIT Bombay: Quantum Sensing & Metrology.
- T-Hub at IIT Delhi: Quantum Materials & Devices.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- PIB parliamentary answer: says the T-Hubs comprise 14 Technical Groups across 17 States and 2 UTs.
- Quantum communication is strategically relevant for secure networks and national-security communication contexts.
- Quantum sensing/metrology can matter for navigation, clocks, precision instrumentation and advanced measurement ecosystems.
- Quantum materials/devices are foundational because hardware dependence can otherwise cripple mission ambition.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.